SEMINAR REPORT

ON

Trusted Execution Environments (TEEs) for Secure Cloud Computing - Performance & Security Trade-offs

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CERTIFICATE

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To my satisfaction and submitted the same during the academic year 2025 - 2026 towards the partial fulfillment of degree of Bachelor of Technology in School of Computer Engineering & Technology under Dr. Vishwanath Karad MIT- World Peace University, Pune.

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ABBREVIATIONS

TEE	Trusted Execution Environment
SGX	Software Guard Extensions
TDX	Trust Domain Extensions
SEV	Secure Encrypted Virtualization
SEV-SNP	Secure Encrypted Virtualization – Secure Nested Paging
VM	Virtual Machine
OS	Operating System
CPU	Central Processing Unit
GPU	Graphics Processing Unit
EPC	Enclave Page Cache
TCB	Trusted Computing Base
I/O	Input / Output

API	Application Programming Interface
ML	Machine Learning
AI	Artificial Intelligence
LLM	Large Language Model
CSP	Cloud Service Provider
TLS	Transport Layer Security
AES	Advanced Encryption Standard
RAM	Random Access Memory
HPC	High Performance Computing
DoS	Denial of Service
VM-Exit	Virtual Machine Exit (transition from VM to hypervisor)
LibOS	Library Operating System
RPS	Requests Per Second

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ABSTRACT

The rapid expansion of cloud computing has transformed the way data is stored, processed, and accessed across industries such as healthcare, finance, artificial intelligence, and scientific research. While cloud platforms offer scalability, flexibility, and cost efficiency, they also introduce significant concerns related to data security and privacy. In traditional cloud environments, sensitive data is protected using encryption mechanisms during storage and transmission; however, during computation, the data must be decrypted, making it vulnerable to attacks from untrusted system components such as the operating system, hypervisor, or even privileged insiders. This fundamental limitation has led to the emergence of Trusted Execution Environments (TEEs) as a promising solution for secure computation in untrusted environments.

Trusted Execution Environments provide a hardware-enforced secure region within a processor where code and data can be executed in isolation. This ensures that sensitive information remains protected even if the rest of the system is compromised. Modern TEE technologies such as Intel Software Guard Extensions (SGX), Intel Trust Domain Extensions (TDX), and AMD Secure Encrypted Virtualization with Secure Nested Paging (SEV-SNP) have gained significant attention in the domain of confidential computing. These technologies differ in their architectural design, level of isolation, scalability, and performance characteristics, making it essential to study their behavior in real-world cloud workloads.

Despite their strong security guarantees, TEEs introduce several practical challenges, particularly in terms of performance overhead, memory limitations, and system scalability. Operations such as enclave transitions, encrypted memory access, and restricted system calls often lead to increased latency and reduced throughput. Furthermore, different TEE architectures exhibit varying trade-offs between performance and security, depending on the nature of the workload. For instance, applications that are memory intensive or I/O bound may experience significant degradation when executed inside secure enclaves. Therefore, understanding and evaluating the balance between security benefits and performance costs is crucial for the effective deployment of TEEs in cloud environments.

In addition to the theoretical analysis, a practical implementation has been developed in the form of a simulation based performance dashboard using Python and Streamlit. The implemented system models and compares the behavior of SGX, TDX, and SEV-SNP under different workload conditions, including variations in dataset size, concurrency levels, and memory pressure. The dashboard computes key performance metrics such as latency, throughput, memory overhead, and a combined performance index, along with a security score for each TEE. Interactive visualizations, including bar charts, scatter plots, and scalability graphs, are used to clearly illustrate the trade-offs between performance and security.

The experimental analysis demonstrates that no single TEE technology is universally optimal. Intel SGX provides the highest level of security through fine grained enclave isolation but suffers from higher latency and memory overhead. Intel TDX offers a balanced approach by providing virtual machine-level isolation with moderate performance impact. AMD SEV-SNP, on the other hand, delivers better performance and scalability due to its memory encryption-based design but offers comparatively less granular isolation. These findings highlight the importance of selecting TEE solutions based on specific application requirements rather than relying on a one-size-fits-all approach.

Overall, this study emphasizes that while Trusted Execution Environments play a crucial role in enabling secure cloud computing, their adoption must be carefully evaluated in the context of performance constraints and workload characteristics. The combination of theoretical insights and practical simulation presented in this report contributes to a deeper understanding of confidential computing systems and provides a foundation for future research and optimization in this domain.

Keywords : Trusted Execution Environment (TEE), Intel SGX, Intel TDX, AMD SEV-SNP, Confidential Computing, Cloud Security, Performance Overhead, Secure Enclaves, Data Privacy, Streamlit Dashboard, Performance Analysis\

CHAPTER 1: INTRODUCTION

1. INTRODUCTION

Cloud computing has become one of the most important technological advancements in modern computing, enabling organizations to access scalable, on-demand computing resources over the internet. It has revolutionized industries such as healthcare, finance, artificial intelligence, and big data analytics by reducing infrastructure costs and improving operational efficiency. However, despite its numerous advantages, cloud computing introduces critical concerns related to data security and privacy, especially when sensitive data is processed on third-party infrastructure.

In conventional cloud environments, data protection is typically achieved through encryption mechanisms. Data is encrypted while being stored (encryption-at-rest) and while being transmitted across networks (encryption-in-transit). However, a significant security gap exists during data processing, where the data must be decrypted and is therefore exposed in plaintext within system memory. This exposure makes it vulnerable to attacks from malicious software, compromised operating systems, hypervisors, or even insider threats. As a result, ensuring secure computation in untrusted environments has become a major research challenge.

To address this issue, Trusted Execution Environments (TEEs) have emerged as a powerful solution for enabling secure and confidential computation. A TEE is a secure area within a processor that provides isolation and protection for code and data during execution. It ensures that sensitive operations are performed in a trusted and protected environment, even if the underlying system software is compromised. This concept forms the foundation of confidential computing, a rapidly growing paradigm in cloud security.

Several TEE technologies have been developed by major hardware vendors to support secure cloud computing. Intel Software Guard Extensions (SGX) provides application-level secure

enclaves that isolate specific portions of code and data. Intel Trust Domain Extensions (TDX) extends this concept to the virtual machine level, allowing entire VMs to run in a protected environment. AMD Secure Encrypted Virtualization (SEV) and its enhanced version SEV-SNP provide memory encryption for virtual machines, ensuring that data remains protected from unauthorized access. Each of these technologies offers unique advantages and limitations in terms of performance, scalability, and security.

Although TEEs significantly enhance security, they also introduce practical challenges that impact system performance. Operations within secure environments often involve additional computational overhead due to encryption, context switching, and restricted access to system resources. These factors can lead to increased latency, reduced throughput, and higher memory consumption. Furthermore, different TEE architectures behave differently under varying workloads, making it necessary to carefully analyze their performance characteristics.

With the growing adoption of cloud-based applications such as artificial intelligence, data analytics, and bioinformatics, understanding the performance-security trade-offs of TEEs has become increasingly important. Workloads such as machine learning inference, database queries, and large scale data processing require both high performance and strong security guarantees. Therefore, selecting an appropriate TEE technology depends on balancing these conflicting requirements.

In this seminar, a comprehensive study of Trusted Execution Environments is conducted with a focus on evaluating their performance and security trade-offs in cloud computing environments. In addition to reviewing recent research contributions in this field, a practical simulation-based approach is implemented to visualize and analyze the behavior of different TEE technologies under varying conditions. A Streamlit-based dashboard is developed to simulate workloads and compare key performance metrics such as latency, throughput, memory overhead, and overall efficiency.

The proposed implementation enables users to explore how different TEEs respond to changes in workload size, concurrency levels, and system constraints. It provides a clear understanding of how security mechanisms influence system performance and helps identify the most suitable TEE for a given application scenario. This combination of theoretical analysis and practical experimentation provides a holistic understanding of confidential computing systems.

1.1 Problem Statement

Despite the advancements in cloud security, protecting data during computation remains a significant challenge. Traditional security approaches fail to ensure confidentiality when data is actively being processed. Although Trusted Execution Environments provide a solution, they introduce performance overheads and scalability limitations that affect system efficiency.

The key problem addressed in this seminar is:

"How to analyze and evaluate the trade-offs between security and performance in Trusted Execution Environments for cloud computing applications?"

Understanding these trade-offs is essential for selecting appropriate TEE technologies for different workloads and optimizing their deployment in real-world scenarios.

1.2 Motivation

The motivation for this study arises from the increasing reliance on cloud computing for handling sensitive data across various domains. Applications such as healthcare systems, financial transactions, and AI-based services require strong data protection mechanisms to ensure privacy and compliance with regulations.

At the same time, these applications demand high performance and low latency, making it challenging to adopt security solutions that introduce significant overhead. The need for secure yet efficient computation has driven the development of TEEs and confidential computing technologies.

Additionally, recent research has shown that different TEE architectures exhibit varying levels of performance degradation depending on workload characteristics. This creates a need for tools and frameworks that can help analyze and visualize these trade-offs in a simplified and understandable manner.

The development of a simulation-based dashboard in this seminar is motivated by the need to bridge the gap between theoretical research and practical understanding, allowing users to explore TEE behavior interactively.

1.3 Objectives

The main objectives of this seminar are as follows:

- To study the concept of Trusted Execution Environments and their role in secure cloud computing.
- To analyze different TEE technologies such as Intel SGX, Intel TDX, and AMD SEV-SNP.
- To evaluate the performance overheads associated with TEE-based execution.
- To understand the trade-offs between security and system performance in cloud environments.
- To develop a simulation-based dashboard for comparing TEE performance under different workloads.
- To visualize key performance metrics such as latency, throughput, and memory overhead.
- To provide insights into selecting suitable TEE technologies for real-world applications.

1.4 Scope of the Report

This seminar report focuses on the study and analysis of Trusted Execution Environments in the context of cloud computing. It covers both theoretical and practical aspects of TEE technologies, including their architecture, working principles, advantages, limitations, and performance characteristics.

The report also includes a simulation-based implementation that demonstrates how different TEEs behave under varying workload conditions. However, the study is limited to software based simulation and does not involve direct deployment on hardware-supported TEE platforms. Future extensions of this work may include real world implementation using cloud platforms that support confidential computing, such as Microsoft Azure Confidential VMs or AWS Nitro Enclaves.

1.5 Organization of the Report

The report is organized as follows:

- **Chapter 1** introduces the concept of Trusted Execution Environments, outlines the problem statement, motivation, objectives, and scope of the study.
- **Chapter 2** presents a detailed literature survey of recent research work related to TEE technologies and confidential computing.
- **Chapter 3** describes the design and architecture of TEE systems and explains the working of different technologies.
- **Chapter 4** discusses the implementation of the simulation dashboard, including methodology, tools, and system workflow.
- **Chapter 5** presents the results and analysis of the simulation, highlighting performance and security trade-offs.
- **Chapter 6** concludes the report and provides insights into future research directions.

CHAPTER 2: LITERATURE SURVEY

2. LITERATURE SURVEY

With the rapid evolution of cloud computing and increasing concerns regarding data privacy, Trusted Execution Environments (TEEs) have gained significant attention as a key solution for secure computation. Researchers have extensively studied TEE technologies to evaluate their performance, scalability, and security trade-offs. This chapter presents a comprehensive review of existing research work, categorizing contributions into performance evaluation, architectural advancements, cloud integration, and security challenges.

2.1 Performance Evaluation of TEE Technologies

Performance evaluation is one of the most critical aspects of TEE research, as security mechanisms often introduce computational overhead. Several studies have focused on benchmarking different TEE technologies to understand their impact on latency, throughput, and memory usage.

Brescia et al. (2026) conducted a detailed experimental study titled *"A Comprehensive Performance Evaluation of TEEs for Confidential DNA Alignment"*. The research evaluated Intel SGX, Intel TDX, and AMD SEV-SNP using real-world bioinformatics workloads. The results showed that SGX provides strong security through enclave isolation but suffers from higher latency due to enclave transitions and limited memory. TDX demonstrated better scalability compared to SGX, while SEV-SNP showed improved throughput and lower overhead in memory-intensive applications.

Similarly, Coppolino et al. (2025) performed benchmarking in *"An Experimental Evaluation of TEE Technology: Benchmarking SGX, SEV, and TDX"*. Their study analyzed performance across different execution environments and runtimes such as Gramine-SGX and Occlum-SGX. The findings indicated that virtualization-based TEEs like SEV and TDX offer better performance for

large-scale workloads, whereas SGX incurs higher overhead due to its fine-grained isolation model.

Another study by Hoffmann et al. (2025) focused on real-time systems in *"On Real-Time Guarantees in Intel SGX and TDX"*. This research highlighted the challenges of using TEEs in latency-sensitive applications such as edge computing and real-time analytics. It concluded that while TDX performs better than SGX in terms of predictability, both technologies require optimization to meet strict real-time constraints.

2.2 Architectural Advancements in TEEs

To address performance and scalability limitations, researchers have proposed new architectural designs and improvements to existing TEE frameworks.

Sunny et al. (2025) introduced *"SecScale: A Scalable and Secure Trusted Execution Environment for Servers"*, which focuses on improving scalability in server environments. The proposed architecture optimizes resource allocation and reduces overhead associated with enclave execution. The study demonstrates that efficient resource management can significantly improve TEE performance without compromising security.

Lu et al. (2026) proposed a novel container-based approach in *"Arca: A Lightweight Confidential Container Architecture for Cloud-Native Environments"*. This work focuses on reducing the Trusted Computing Base (TCB) and improving performance for cloud-native applications. The proposed system integrates TEEs with container technologies, enabling secure and efficient deployment of microservices.

Another important contribution is presented in *"The Road to Trust: Building Enclaves within Confidential VMs"* (2024), where the concept of nested enclaves is introduced. This approach combines the benefits of enclave-based and virtualization-based TEEs, allowing secure execution within confidential virtual machines. The study demonstrates improved flexibility and security but highlights additional overhead introduced by layered architectures.

2.3 TEE Integration in Cloud Systems

The integration of TEEs into cloud infrastructure is a major area of research, focusing on real-world applications such as storage systems, databases, and AI workloads.

Burke et al. (2025) explored TEE-based cloud storage in *"Securing Cloud File Systems With Trusted Execution"*. The study demonstrated how TEEs can be used to protect data in distributed file systems. The results showed that TEE integration enhances data security but introduces overhead in file access operations, particularly for I/O-intensive workloads.

Zhang et al. (2025) investigated AI workloads in *"Evaluating the Performance of the DeepSeek Model in Confidential Computing Environment"*. This research analyzed the performance of large language models (LLMs) within TEE-enabled environments. The findings revealed that secure execution significantly impacts inference speed, highlighting the need for optimization techniques for AI-based applications.

Additionally, research on confidential computing platforms shows that major cloud providers such as AWS, Azure, and Google Cloud are adopting TEE technologies to enable secure data processing. These platforms provide confidential virtual machines and secure enclaves, allowing organizations to run sensitive workloads in untrusted environments.

2.4 Security Challenges and Vulnerabilities

While TEEs provide strong isolation and protection, they are not completely immune to security threats. Several studies have identified vulnerabilities and limitations in TEE-based systems.

The Systematization of Knowledge (SoK) paper *"Understanding Design Choices and Pitfalls of Trusted Execution Environments"* (2024) provides a comprehensive analysis of TEE architectures. The study highlights various security challenges, including side-channel attacks, memory access leakage, and interface vulnerabilities. It emphasizes that while TEEs reduce the attack surface, they do not eliminate all risks.

Similarly, research on confidential computing surveys (IET, 2024) discusses the limitations of current TEE implementations, including susceptibility to cache-based attacks and speculative execution vulnerabilities. These issues highlight the importance of continuous research in improving TEE security mechanisms.

Another important aspect is the trade-off between security and usability. Stronger security mechanisms often lead to reduced flexibility and increased complexity in application development. This makes it challenging for developers to adopt TEE technologies in large-scale systems.

Table 1: Literature review Summary

Sr. No.	Author(s) & Year	Title of Paper	TEE Technology Used	Key Contribution	Findings / Limitations
1	Brescia et al. (2026)	A comprehensive performance evaluation of TEEs for confidential DNA alignment	SGX, TDX, SEV-SNP	Real-world benchmarking using bioinformatics workloads	SGX has high security but high latency; SEV-SNP offers better throughput
2	Coppolino et al. (2025)	An experimental evaluation of TEE technology: Benchmarking SGX, SEV, and TDX	SGX, SEV, TDX	Comparative performance analysis across TEEs and runtimes	SGX has higher overhead; SEV and TDX are more scalable

3	Sunny et al. (2025)	SecScale: A scalable and secure trusted execution environment for servers	Custom TEE Architecture	Proposed scalable server-side TEE design	Improves scalability but increases system complexity
4	Burke et al. (2025)	Securing Cloud File Systems With Trusted Execution	SGX-based system	Integration of TEE in cloud file systems	Secure storage but increased I/O latency
5	Hoffmann et al. (2025)	On Real-Time Guarantees in Intel SGX and TDX	SGX, TDX	Real-time performance evaluation in edge/cloud systems	SGX struggles with latency; TDX performs better but needs optimization
6	Lu et al. (2026)	Arca: A Lightweight Confidential Container Architecture	SGX, TDX	Confidential container-based architecture	Reduces TCB but still has overhead
7	Lind et al. (2024)	The Road to Trust: Building Enclaves within Confidential VMs	SGX + VM	Nested enclave architecture	Improves flexibility but adds overhead
8	Coppolino et al. (2024)	Experimental evaluation of TEE evolution	SGX, SEV, TDX	Study of evolution of TEE architectures	Shows trade-offs between security and performance

9	Zhang et al. (2025)	Evaluating DeepSeek Model in Confidential Computing	TEE-based AI	AI/LLM performance in TEE environment	Significant performance drop in inference
10	Sabt et al. (2024)	Survey of research on confidential computing	SGX, SEV, TDX	Comprehensive survey of TEE technologies	Highlights security risks like side-channel attacks
11	Wang et al. (2024)	SoK: Understanding design choices and pitfalls of TEEs	Multiple TEEs	Systematization of TEE design	Identifies architectural weaknesses
12	Anasuri (2024)	Confidential Computing Using TEEs	General TEE	Overview of confidential computing	High-level, lacks experimental validation
13	Gu et al. (2020)	Enabling rack-scale confidential computing	Heterogeneous TEEs	Large-scale cloud deployment study	Shows scalability challenges

CHAPTER 3: DESIGN AND ARCHITECTURE

3. DESIGN AND ARCHITECTURE

Trusted Execution Environments are designed to provide secure execution of sensitive applications in potentially untrusted systems. In real-world cloud environments, TEEs ensure that data and computations remain protected inside hardware isolated regions such as enclaves or confidential virtual machines. However, due to hardware and deployment limitations, direct implementation of TEEs is not always feasible in an academic setup.

To address this, the proposed system adopts a simulation-based approach that models the behavior of TEEs. While the system does not execute code within actual hardware enclaves, it closely represents the workflow, performance characteristics, and security implications observed in real TEE enabled systems. This design allows for better understanding and visualization of how TEEs function in practice.

3.1 Overview of TEE Architecture

In real-world systems, the architecture of a Trusted Execution Environment is based on a clear separation between trusted and untrusted components. The trusted environment operates as a secure execution region where sensitive computations are performed, while the untrusted environment includes the operating system, applications, and other system-level components.

This separation ensures that even if the operating system is compromised, the data inside the trusted region remains protected. Communication between these two environments is carefully controlled, typically involving secure transitions that enforce strict access policies.

In the proposed simulation model, this architecture is conceptually replicated. The system distinguishes between input handling, secure computation, and output generation stages. Although actual hardware isolation is not implemented, the model simulates the behavior of

trusted and untrusted environments through controlled data processing and performance modeling.

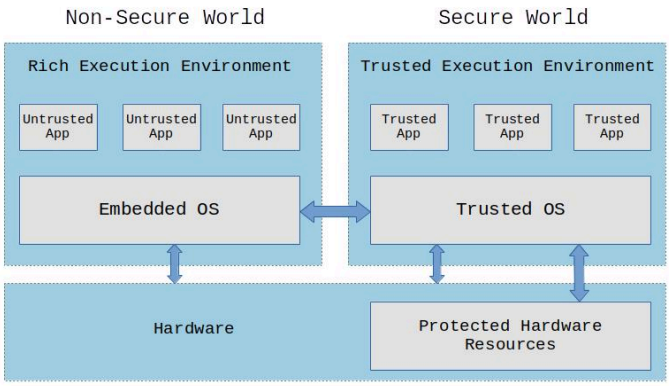


Fig 1: TEE Environment

3.2 Working Principle of TEEs

The functioning of TEEs in real systems is governed by several key principles that ensure secure execution. These principles are also reflected in the simulation model to approximate real-world behavior.

1. **Isolation:**
   In actual TEEs, sensitive code is executed in an isolated region such as an enclave or secure virtual machine. This prevents unauthorized access from the operating system. In the simulation, this isolation is represented by separating the computation logic within a dedicated module that processes inputs independently.
2. **Confidentiality:**
   Real TEEs use encryption and access control to protect data. The simulation models this by assigning security levels to different TEE technologies, representing their relative ability to protect data.
3. **Integrity:**
   TEEs ensure that data and code are not modified during execution. In the proposed

system, integrity is represented through controlled processing steps that maintain consistency of computed results.

4. **Remote Attestation:**
   In real systems, attestation verifies the authenticity of the execution environment. While this feature is not implemented directly, the simulation reflects its importance by incorporating predefined trusted configurations for each TEE.

3.3 Types of TEE Technologies

The simulation model focuses on three major TEE technologies that are widely used in cloud computing. Each of these technologies is represented in the system with its respective performance and security characteristics.

3.3.1 Intel SGX (Software Guard Extensions)

In real-world implementations, Intel SGX provides application-level security by creating enclaves within the processor. These enclaves execute sensitive portions of code in isolation, ensuring a high level of protection. In the simulation model, SGX is represented as a highly secure but performance-intensive environment. The model assigns higher latency and memory overhead to SGX to reflect the cost of enclave transitions and limited enclave memory. This representation helps demonstrate how strong security can impact system performance.

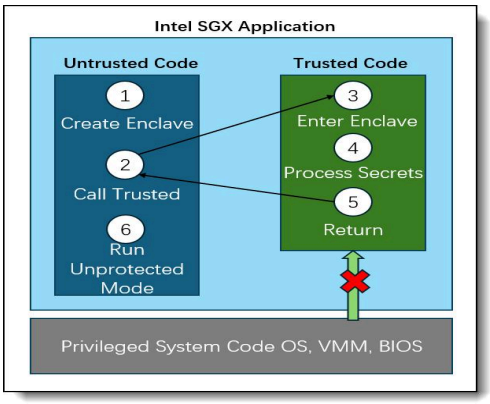


Fig 2 : SGX Application Architecture

3.3.2 Intel TDX (Trust Domain Extensions)

Intel TDX extends the concept of secure execution to entire virtual machines. This allows cloud workloads to run in protected environments without modifying application code significantly. In the simulation, TDX is modeled as a balanced approach that provides moderate security with improved performance compared to SGX. The system reflects its scalability advantages by assigning better throughput values, making it suitable for larger workloads.

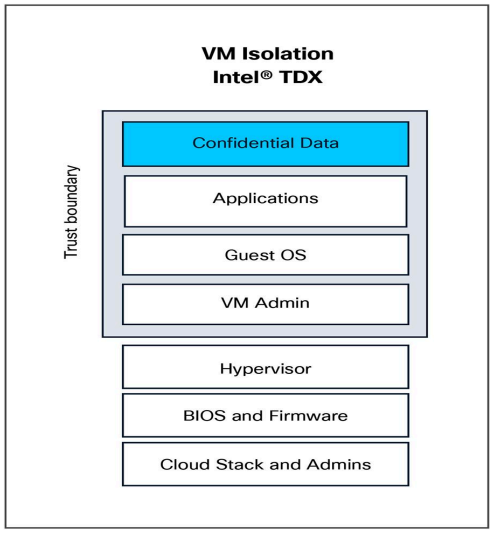


Fig 3: TDX diagram

3.3.3 AMD SEV-SNP (Secure Encrypted Virtualization)

AMD SEV-SNP protects virtual machines by encrypting memory, ensuring that sensitive data cannot be accessed by unauthorized components. Within the simulation model, SEV-SNP is represented as a performance efficient solution with relatively lower latency and higher throughput. However, it is assigned a slightly lower security level compared to SGX to reflect its less granular isolation..

3.4 System Architecture of the Proposed Model

The proposed system is designed as a simulation framework that mimics the workflow of TEE-based execution in cloud environments. Although the implementation is software-based, it follows the same logical stages as a real TEE system.

The system begins with a user interface developed using Streamlit, where users provide input parameters such as workload type and concurrency level. These inputs represent real application scenarios that would typically be executed inside a secure enclave or confidential virtual machine.

The input data is then processed by a dedicated module that prepares it for execution. In real systems, this stage corresponds to preparing data before entering the secure execution environment. The core of the system is the simulation engine, which acts as a virtual representation of the TEE. In actual implementations, this would correspond to an enclave or secure VM where computation takes place. The simulation engine calculates performance metrics such as latency, throughput, and memory overhead based on predefined models.

Finally, the results are passed to the visualization module, which generates graphical outputs. In real-world scenarios, this stage represents exiting the secure environment and returning results to the application layer.

3.5 System Flow

The working of the system follows a structured sequence that mirrors real TEE operations. Initially, the user provides input parameters through the interface. These parameters are then processed and forwarded to the simulation engine.

The simulation engine performs computations that represent secure execution inside a TEE. After processing, the results are sent to the visualization module, where they are presented in graphical form.

This flow ensures that the simulation accurately represents the lifecycle of secure computation, from input processing to result generation.

User Input

↓

Streamlit Interface

↓

Data Processing Module

↓

Simulation Engine (Represents Enclave / Secure VM)

↓

Visualization Module

↓

Output Graphs

3.6 Mathematical Model

The simulation uses simplified mathematical expressions to estimate system behavior. Latency is calculated by applying an overhead factor to a base execution time, representing the cost of secure execution. Throughput is derived based on the number of processed requests over time.

A performance index is computed to provide an overall measure of system efficiency. Although these calculations are simplified, they effectively capture the relative differences between TEE technologies.

- Latency = Base latency × overhead
- Throughput = Requests per second
- Performance Index = Throughput / Latency

3.7 Advantages and Limitations of the Proposed Design

The proposed design provides an accessible way to study TEE behavior without requiring specialized hardware. It allows users to understand how different TEE technologies perform under varying conditions. Additionally, the interactive nature of the system makes it suitable for educational and research purposes.

The system is based on simulation rather than real hardware execution. As a result, certain low-level behaviors and optimizations are not fully captured. Additionally, the model is limited to a few selected TEE technologies and does not include all possible configurations.

3.8 Summary

This chapter presented the design and architecture of Trusted Execution Environments and described how their behavior is represented in the proposed simulation model. Although the implementation is software based, it effectively captures the key aspects of secure computation in TEEs. The architecture highlights the relationship between security mechanisms and performance overhead, providing valuable insights into the trade-offs involved.

CHAPTER 4: IMPLEMENTATION

4. IMPLEMENTATION

This chapter describes the practical implementation of the proposed system for analyzing the performance and security trade-offs of Trusted Execution Environments. The system is developed as a simulation based dashboard using Python and Streamlit. Although the implementation does not use actual hardware level TEEs, it closely models the behavior of real-world secure execution environments such as SGX, TDX, and SEV-SNP.

4.1 Development Environment

The implementation is carried out using a simple yet effective development environment that supports rapid prototyping and visualization. The system is developed using Python as the primary programming language due to its simplicity and strong support for data processing and visualization. The user interface is created using the Streamlit framework, which allows the development of web-based interactive dashboards with minimal complexity.

The key tools and technologies used in the implementation include:

1. **Visual Studio Code (VS Code):**
   Used as the development environment for writing and managing code.
2. **Python:**
   Used for implementing the simulation logic and data processing.
3. **Streamlit:**
   Used for building the interactive user interface.
4. **Pandas:**
   Used for handling and processing structured data.
5. **Matplotlib / Plotly:**
   Used for generating graphical visualizations.

4.2 System Implementation Approach

The implementation follows a simulation-based approach where the behavior of TEE technologies is modeled using predefined parameters derived from research studies. Instead of executing actual secure computations inside hardware enclaves, the system approximates their behavior using mathematical calculations and visualization techniques.

In real-world systems, applications are executed within secure environments such as enclaves or confidential virtual machines. These environments enforce isolation and security while introducing certain performance overheads. In the proposed model, this behavior is simulated by assigning different performance characteristics to each TEE technology.

The implementation focuses on capturing the following aspects:

1. Latency introduced by secure execution
2. Throughput under varying workloads
3. Memory overhead associated with TEE usage
4. Relative security levels of different technologies

4.3 Dataset and Parameters

The system uses a predefined dataset that represents the performance characteristics of different TEE technologies. These values are based on observations from research literature and are used to simulate realistic behavior.

The dataset includes parameters such as:

1. **TEE Type:** Represents SGX, TDX, and SEV-SNP
2. **Latency:** Indicates the delay introduced during execution
3. **Throughput:** Represents the number of requests processed per unit time
4. **Memory Overhead:** Indicates additional memory usage due to security mechanisms
5. **Security Level:** Represents the relative strength of protection

4.4 Implementation of Simulation Engine

The simulation engine is the core component of the system. It is responsible for modeling the behavior of TEE technologies and computing performance metrics based on user input.

In real-world systems, secure execution takes place within enclaves or protected virtual machines. These environments introduce overhead due to encryption, context switching, and restricted access to resources. The simulation engine replicates this behavior by applying predefined overhead factors to the input parameters.

The working of the simulation engine can be described as follows:

1. The user selects workload parameters such as dataset size and concurrency level.
2. The system adjusts base performance values based on these inputs.
3. Overhead factors specific to each TEE technology are applied.
4. Final performance metrics such as latency and throughput are calculated.

4.5 User Interface Implementation

The user interface is implemented using Streamlit, which provides an interactive platform for user input and result visualization. The interface is designed to be simple and intuitive, allowing users to easily explore different scenarios.

The interface includes input controls such as dropdown menus and sliders, which allow users to select workload parameters. Once the inputs are provided, the system processes the data and displays the results in real time.

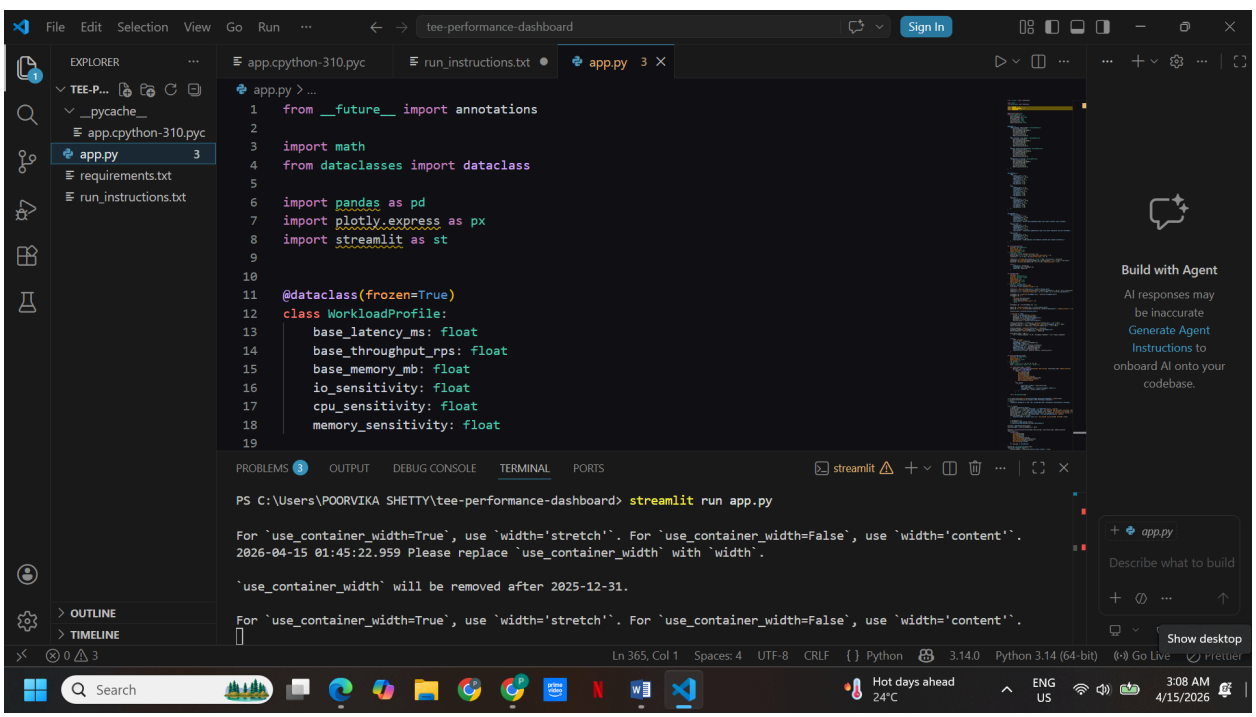
The output is presented in the form of tables and graphs, making it easier to compare different TEE technologies. This interactive design enhances user understanding and makes the system more engaging.

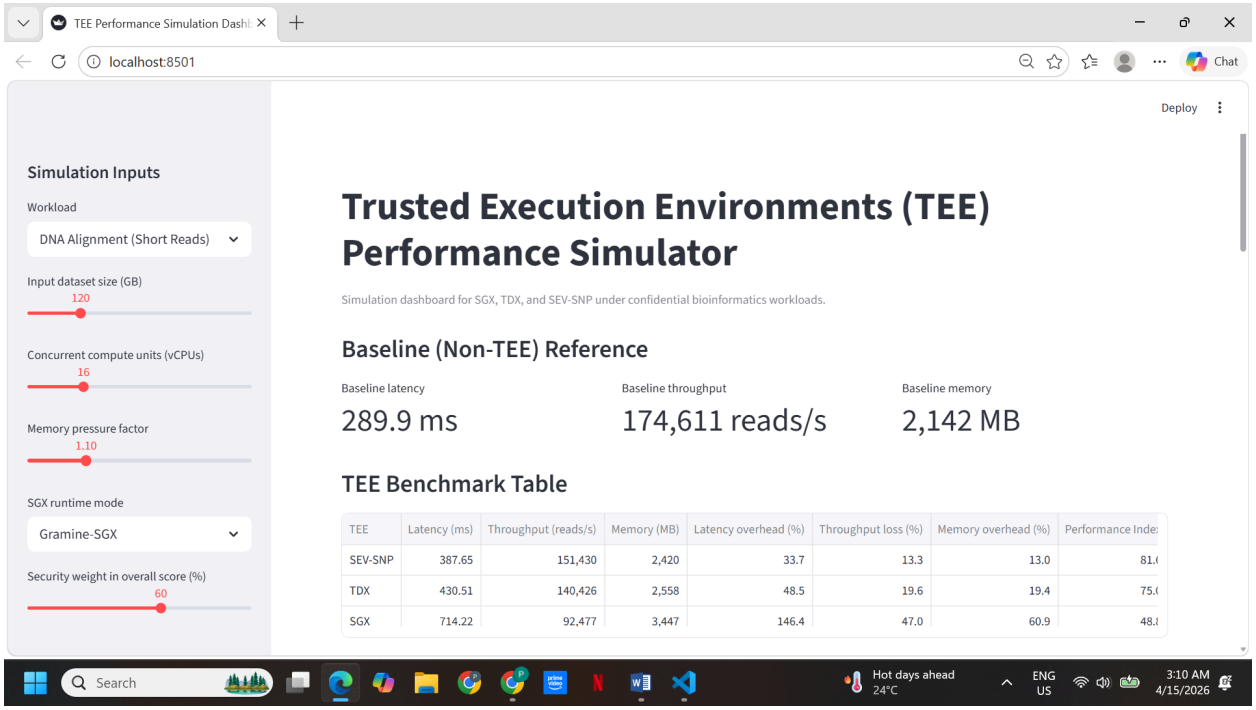
4.6 Visualization of Results

Visualization plays a crucial role in understanding performance trade-offs. The system generates multiple types of graphs to represent different aspects of TEE behavior.

1. **Latency Comparison Graph:**
   Shows how execution delay varies across different TEE technologies.
2. **Throughput Comparison Graph:**
   Displays the processing capacity of each TEE under different conditions.
3. **Memory Overhead Graph:**
   Illustrates additional resource consumption due to secure execution.
4. **Performance Comparison Graph:**
   Provides an overall comparison of system efficiency.

4.7 Implementation and Visualization Screenshots:




TEE Performance Simulation Dashl
localhost:8501
Chat
Deploy
Simulation Inputs
Workload
DNA Alignment (Short Reads)
Input dataset size (GB)
120
Concurrent compute units (vCPUs)
16
Memory pressure factor
1.10
SGX runtime mode
Gramine-SGX
Security weight in overall score (%)
60
Trusted Execution Environments (TEE) Performance Simulator
Simulation dashboard for SGX, TDX, and SEV-SNP under confidential bioinformatics workloads.
Baseline (Non-TEE) Reference
Baseline latency
289.9 ms
Baseline throughput
174,611 reads/s
Baseline memory
2,142 MB
TEE Benchmark Table
TEE | Latency (ms) | Throughput (reads/s) | Memory (MB) | Latency overhead (%) | Throughput loss (%) | Memory overhead (%) | Performance Inde
SEV-SNP | 387.65 | 151,430 | 2,420 | 33.7 | 13.3 | 13.0 | 81.
TDX | 430.51 | 140,426 | 2,558 | 48.5 | 19.6 | 19.4 | 75.
SGX | 714.22 | 92,477 | 3,447 | 146.4 | 47.0 | 60.9 | 48.
Search
Hot days ahead
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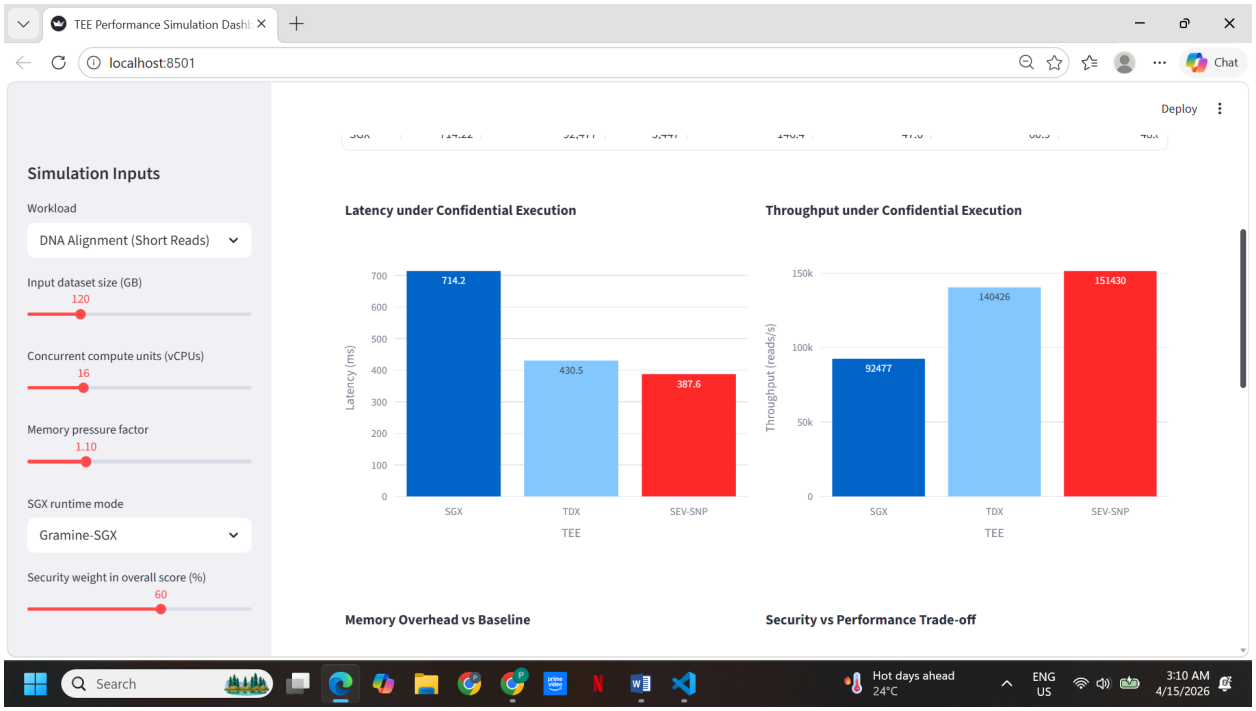
3:10 AM
4/15/2026


TEE Performance Simulation Dashl
localhost:8501
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Simulation Inputs
Workload
DNA Alignment (Short Reads)
Input dataset size (GB)
120
Concurrent compute units (vCPUs)
16
Memory pressure factor
1.10
SGX runtime mode
Gramine-SGX
Security weight in overall score (%)
60
Latency under Confidential Execution
Latency (ms)
714.2
430.5
387.6
SGX
TDX
SEV-SNP
TEE
Throughput under Confidential Execution
Throughput (reads/s)
92477
140426
151430
Memory Overhead vs Baseline
Security vs Performance Trade-off
Search
Hot days ahead
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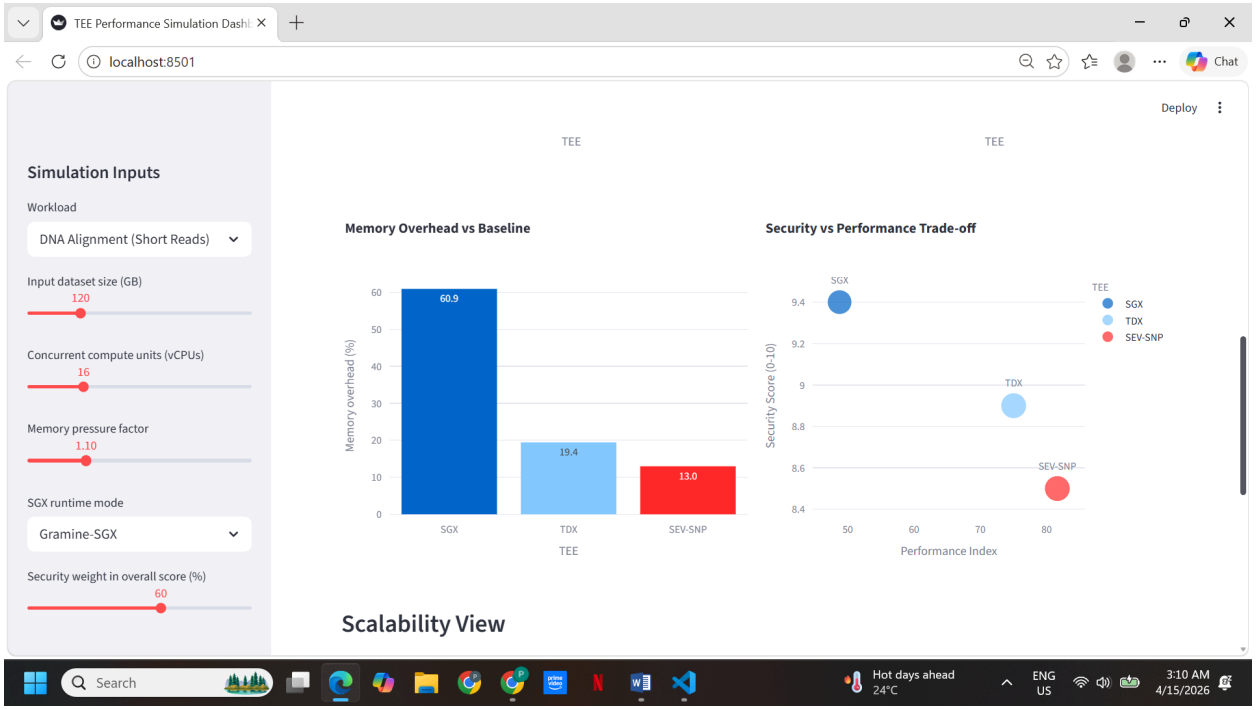
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Workload
DNA Alignment (Short Reads)
Input dataset size (GB)
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Concurrent compute units (vCPUs)
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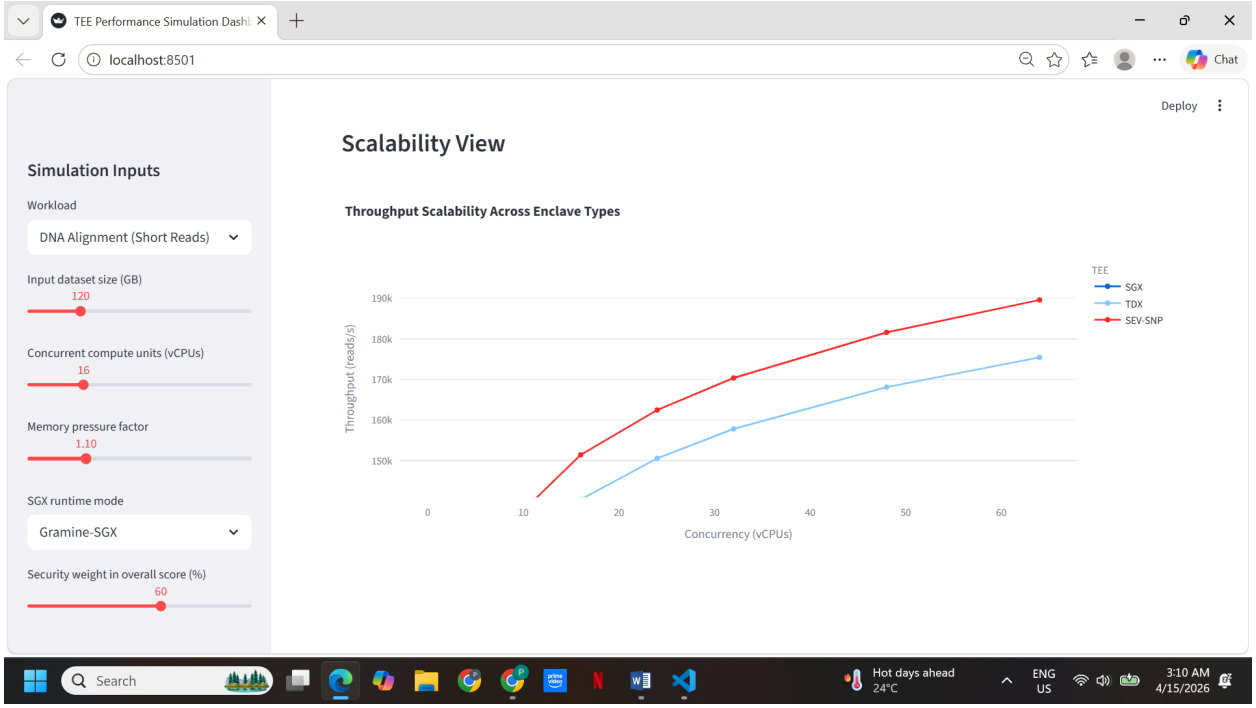
Memory pressure factor
1.10
SGX runtime mode
Gramine-SGX
Security weight in overall score (%)
60
TEE
Memory Overhead vs Baseline
Memory overhead (%)
60.9
19.4
13.0
SGX
TDX
SEV-SNP
TEE
Security vs Performance Trade-off
Security Score (0-10)
Performance Index
Scalability View
Search
Hot days ahead
24°C
ENG
US
3:10 AM
4/15/2026


TEE Performance Simulation Dashb
localhost:8501
Chat
Deploy
Scalability View
Simulation Inputs
Workload
DNA Alignment (Short Reads)
Input dataset size (GB)
120
Concurrent compute units (vCPUs)
16
Memory pressure factor
1.10
SGX runtime mode
Gramine-SGX
Security weight in overall score (%)
60
Throughput Scalability Across Enclave Types
Throughput (reads/s)
Concurrency (vCPUs)
TEE
SGX
TDX
SEV-SNP
Search
Hot days ahead
24°C
ENG
US
3:10 AM
4/15/2026

4.8 Mapping Simulation to Real-World Execution

Although the system is based on simulation, it closely reflects real-world TEE execution. In actual implementations, the workflow involves entering a secure environment, performing computation, and exiting with results.

The simulation follows a similar structure, where user input represents application data, the simulation engine represents the secure execution environment, and the visualization module represents the output stage.

This mapping ensures that the system provides a realistic understanding of how TEEs operate, even without direct hardware implementation.

4.9 Summary

This chapter presented the implementation of a simulation-based system for analyzing Trusted Execution Environments. The system uses Python and Streamlit to create an interactive dashboard that models the performance and security characteristics of SGX, TDX, and SEV-SNP.

Although the implementation does not involve actual hardware TEEs, it effectively simulates their behavior and provides valuable insights into their performance trade-offs. The system demonstrates how secure computation impacts system efficiency and highlights the importance of selecting appropriate TEE technologies based on application requirements.

CHAPTER 5: RESULTS AND ANALYSIS

5. RESULTS AND ANALYSIS

This chapter presents the results obtained from the simulation-based Trusted Execution Environment (TEE) performance dashboard and provides a detailed analysis of the observed behavior. The results are derived by simulating different workload conditions and comparing the performance of Intel SGX, Intel TDX, and AMD SEV-SNP in terms of latency, throughput, memory overhead, scalability, and overall efficiency.

Although the system is based on simulation, the results closely reflect real-world trends observed in TEE based cloud environments. The analysis highlights the trade-offs between security and performance, which is a critical factor in selecting appropriate TEE technologies.

5.1 Baseline vs TEE Performance

The baseline represents a non-secure execution environment where no TEE mechanisms are applied. It provides a reference point for evaluating the performance overhead introduced by TEEs.

The simulation results indicate that all TEE technologies introduce additional latency and reduce throughput compared to the baseline system. This is expected, as secure execution requires encryption, isolation, and controlled access to system resources. Among the evaluated technologies, SGX shows the highest deviation from the baseline, indicating significant overhead. In contrast, SEV-SNP demonstrates relatively lower deviation, suggesting better performance efficiency.

This comparison highlights the fundamental trade-off between security and performance, where stronger isolation often leads to increased computational cost.

5.2 Latency Analysis

Latency represents the time required to complete a given task under secure execution. The results clearly show that SGX exhibits the highest latency among the three technologies. This is primarily due to the overhead associated with enclave transitions and limited secure memory.

TDX shows moderate latency, indicating improved efficiency compared to SGX. Since TDX operates at the virtual machine level, it reduces the frequency of secure transitions, resulting in better performance.

SEV-SNP demonstrates the lowest latency, making it the most efficient option in terms of execution time. This is because memory encryption-based approaches introduce less overhead compared to enclave-based isolation.

Overall, the latency analysis suggests that applications requiring low response time may benefit more from SEV-SNP or TDX rather than SGX.

Table 2: Latency Analysis

TEE Type	Latency (ms)	Relative Performance
SGX	714.22	High latency due to enclave transitions
TDX	430.51	Moderate latency with VM-level isolation
SEV-SNP	387.65	Lowest latency, efficient execution

5.3 Throughput Analysis

Throughput measures the number of operations or requests processed per unit time. The results indicate that SEV-SNP achieves the highest throughput, followed by TDX and SGX.

The lower throughput observed in SGX can be attributed to its fine-grained security model, which introduces additional overhead for each operation. TDX improves throughput by providing VM-level isolation, reducing the cost of repeated secure transitions.

SEV-SNP, on the other hand, achieves superior throughput due to its efficient memory encryption mechanism, which allows faster data processing without significant interruption.

This analysis shows that for high-performance applications such as data analytics and machine learning, SEV-SNP offers a more suitable solution.

5.4 Memory Overhead Analysis

Memory overhead represents the additional memory required due to secure execution. The results show that SGX has the highest memory overhead, primarily due to the limited size of the Enclave Page Cache (EPC) and the need for secure memory management.

TDX demonstrates moderate memory overhead, as it operates at the virtual machine level and does not impose strict memory limitations like SGX. SEV-SNP shows the lowest memory overhead, making it more suitable for memory-intensive applications.

This analysis highlights that applications with large datasets may face performance issues when executed using SGX, whereas SEV-SNP provides a more scalable solution.

5.5 Security vs Performance Trade-off

One of the most important observations from the simulation is the trade-off between security and performance. SGX provides the highest level of security due to its fine-grained enclave-based

isolation. However, this comes at the cost of increased latency, reduced throughput, and higher memory usage.

TDX offers a balanced approach by providing strong security at the virtual machine level while maintaining moderate performance. SEV-SNP provides better performance and scalability but slightly lower security compared to SGX.

The graphical representation of this trade-off clearly shows that no single TEE technology is optimal for all scenarios. The choice depends on the specific requirements of the application, such as whether security or performance is the primary concern.

Table 3: Security and Performance Trade-offs

TEE Type	Security Level (0–10)	Performance Index	Observation
SGX	9.5	48.8	Highest security, lowest performance
TDX	8.9	75.0	Balanced approach
SEV-SNP	8.5	81.0	Best performance, slightly lower security

5.6 Scalability Analysis

Scalability analysis examines how the system performs as the workload increases. The results indicate that SEV-SNP scales most efficiently with increasing concurrency levels, followed by TDX and SGX.

SGX shows limited scalability due to its enclave-based design and memory constraints. TDX performs better by leveraging virtual machine isolation, allowing it to handle higher workloads more effectively.

SEV-SNP demonstrates the best scalability, as its design allows efficient handling of large workloads with minimal overhead. This makes it suitable for cloud environments where scalability is a critical requirement.

Table 4: Performance Analysis

Concurrency Level (vCPUs)	SGX Performance	TDX Performance	SEV-SNP Performance
Low (≤10)	Acceptable	Good	Very Good
Medium (10–30)	Moderate	High	Very High
High (≥50)	Limited	Scalable	Highly Scalable

5.7 Comparative Analysis

The overall comparison of TEE technologies reveals distinct characteristics for each approach. SGX is best suited for applications requiring maximum security, such as secure financial transactions or confidential data processing. However, its performance limitations make it less suitable for large-scale applications.

TDX provides a balanced solution that offers both security and performance, making it suitable for general cloud workloads. SEV-SNP, with its superior performance and scalability, is ideal for data-intensive applications such as AI and big data analytics.

This comparative analysis emphasizes the importance of selecting TEE technologies based on application-specific requirements.

Table 5: Comparative Analysis

Parameter	**Best TEE**	**Worst TEE**	**Reason**
Latency	SEV-SNP	SGX	Lower overhead
Throughput	SEV-SNP	SGX	Efficient processing
Memory Usage	SEV-SNP	SGX	Less memory constraint
Security	SGX	SEV-SNP	Strongest isolation
Scalability	SEV-SNP	SGX	Better workload handling

6. CONCLUSION

This study examined Trusted Execution Environments (TEEs) as an effective approach for enabling secure computation in cloud environments. With increasing reliance on cloud platforms for handling sensitive data, ensuring data protection during processing has become a critical requirement. TEEs address this challenge by providing isolated execution environments, thereby improving confidentiality and integrity.

In this work, three major TEE technologies Intel SGX, Intel TDX, and AMD SEV-SNP were analyzed based on their design and performance characteristics. A simulation-based dashboard was developed to model their behavior and evaluate key parameters such as latency, throughput, memory overhead, and overall efficiency. Although the implementation is not hardware-based, it successfully represents real-world TEE operations and highlights their practical implications.

The results indicate a clear trade-off between security and performance. SGX offers the highest level of security due to its enclave based isolation but suffers from higher latency and resource overhead. TDX provides a balanced solution with moderate performance and security, while SEV-SNP demonstrates superior performance and scalability, making it suitable for large-scale applications.

Overall, the study concludes that no single TEE technology is ideal for all scenarios. The selection of an appropriate TEE depends on the specific requirements of the application, particularly the balance between security and performance. TEEs are a promising solution for secure cloud computing, but further improvements are needed to enhance their efficiency and scalability.

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